Combinations. Permutations of a multiset and combinations with repetition

1. Combinations without repetition:

Def. The k-combinations of n objects (without repetition) are the choices of k objects from n distinct objects, where the order does not matter.

Denote: $\binom{n}{k}$ the number of k-combinations of n objects.

Formula: $\binom{n}{k} = \frac{n^{\underline{k}}}{k!} = \frac{n(n-1)\dots(n-k+1)}{k!} = \frac{n!}{(n-k)!\,k!}$

Property: $\binom{n}{k} = \binom{n}{n-k}$

Problems which reduce to combinations:

1) Strictly increasing sequences $= \binom{n}{k}$

2) How many n bit strings have k ones? $= \binom{n}{k}$

3) Lattice paths (p squares horizontally, q squares vertically) $= \binom{p+q}{p}$

Obs. The sum of the binomial coefficients:

$$\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$$

2. Permutations of a multiset:

The general problem: In how many ways can we order n objects of Ω kinds: n_1 identical ones of kind 1, n_2 identical ones of kind 2, ..., n_Ω identical ones of kind Ω ($n_1+n_2+\dots+n_\Omega=n$)

Def. The number of permutations of a multiset is:

$$\binom{n}{n_1, n_2, \dots, n_\Omega} = \frac{n!}{n_1!\, n_2! \dots n_\Omega!}$$

Obs.

$$\mathcal{E}\binom{n}{n_1, n_2, \dots, n_\Omega} = \Omega^n$$

3. Combinations with repetition:

Def. The number of combinations with repetition is:

$$\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k} = \binom{n+k-1}{n-1} = \frac{\overline{n}^{\underline{k}}}{k!}$$

Obs. There are some special values:

$$\left(\!\binom{n}{0}\!\right) = 1, \quad \left(\!\binom{n}{1}\!\right) = n, \quad \left(\!\binom{1}{k}\!\right) = 1$$

Problems that reduce to combination with repetitions:

1) the non-decreasing sequence
2) the solution in $\mathbb{N}$ of the eq. $k_1+\dots+k_n=k$
3) stars and bars

4. Multinomial theorem:

Def. For any $n, \Omega \in \mathbb{N}^*$:

$$(x_1+x_2+\dots+x_\Omega)^n = \mathcal{E}\binom{n}{n_1, \dots, n_\Omega} x_1^{n_1} x_2^{n_2} \dots x_\Omega^{n_\Omega}$$

Obs. The theorem links all three:

- the k-permutations with rep.: the n^m products, before grouping
- the permutation of a multiset: the coefficients
- the combinations with repetition: the nr. of terms in the final form